Sri Lanka Institute of Information Technology



SE2030 - SoftwareEngineering
Year 1, Semester 1- 2025

Lab Submission 2

Group 26-MTR-IT2011-12

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Identified Personas

1. Visitor
2. Customer
3. Sales Staff
4. Operations Staff
5. Finance Staff
6. Administrator

User Stories

1) Persona 1 – Visitor

01. As a visitor, I want to browse apartment listings, so that I can see the available apartments without creating an account.

02. As a visitor, I want to search and filter apartments by location, price, size, and number of bedrooms, so that I can quickly find suitable apartments.

03. As a visitor, I want to view apartment photos, floor plans, facilities, and prices, so that I can understand the details of each apartment.

04. As a visitor, I want to register as a customer, so that I can reserve apartments and access customer services.

2) Persona 2 – Customer

05. As a customer, I want to book an apartment viewing appointment, so that I can visit and inspect an apartment before purchasing it.

06. As a customer, I want to reserve an available apartment online, so that another customer cannot purchase it while I complete the process.

07. As a customer, I want to upload payment evidence, so that the finance staff can verify my payment.

08. As a customer, I want to view my reservations, appointments, payments, and purchase progress, so that I can track the status of my apartment purchase.

3) Persona 3 – Sales Staff

09. As a sales staff member, I want to record customer inquiries, so that I can maintain a complete history of customer communication.

10. As a sales staff member, I want to assign and manage customer leads, so that every interested customer receives proper follow-up.

11. As a sales staff member, I want to approve, reject, or reschedule viewing appointments, so that apartment visits can be properly organized.

12. As a sales staff member, I want to convert a completed reservation into a sale, so that the final apartment purchase can be recorded.

4) Persona 4 – Operations Staff

13. As an operations staff member, I want to add and update apartment projects, so that customers can view accurate project information.

14. As an operations staff member, I want to add apartment units with price, size, floor, bedrooms, and facilities, so that each apartment can be properly listed.

15. As an operations staff member, I want to update apartment status as Available, Reserved, Sold, or Temporarily Blocked, so that the system displays correct availability.

16. As an operations staff member, I want to upload apartment images, floor plans, brochures, and virtual-tour links, so that customers can obtain complete apartment information.

5) Persona 5 – Finance Staff

17. As a finance staff member, I want to view customer payment submissions, so that I can verify reservation and purchase payments.

18. As a finance staff member, I want to approve or reject payment evidence, so that only valid payments are recorded in the system.

19. As a finance staff member, I want to create and manage installment plans, so that customer payment schedules and outstanding balances can be tracked.

20. As a finance staff member, I want to generate payment receipts and financial reports, so that customers and management can obtain accurate financial records.

6) Persona 6 – Administrator

21. As an administrator, I want to manage customer and staff accounts, so that system users can be properly controlled.

22. As an administrator, I want to assign roles and permissions, so that each user can access only authorized system functions.

23. As an administrator, I want to monitor apartments, reservations, payments, appointments, and sales, so that I can supervise the overall system.

24. As an administrator, I want to generate system reports, so that management can make decisions using accurate information.

Scrum Roles

Member	Scrum Role	Responsibility
IT25100393	Product Owner	Maintains and prioritizes the product backlog, clarifies requirements, accepts completed work, and coordinates Payment and Transaction requirements.
IT25103871	Scrum Master	Facilitates Scrum events, monitors sprint progress, removes blockers, and coordinates integration activities.
IT25103860	Developer 1	Develops Apartment Listing Management and apartment search functions.
IT25103887	Developer 2	Develops Viewing Appointment and Reservation Management functions.
IT25102539	Developer 3	Develops Reviews, Notifications and Favorites Management functions.
IT25103131	Developer 4	Develops Apartment Sales and Record Management functions and supports reporting.

Product Backlog Table

User Story ID	Short Description	Priority
01	Browse apartment listings	High
02	Search and filter apartments by location, price, size, and bedrooms	High
03	View apartment details, photos, floor plans, facilities, and prices	High
04	Register as a customer	High
05	Book an apartment viewing appointment	High
06	Reserve an available apartment online	High
07	Upload payment evidence	High
08	View my appointments, reservations, payments, and purchase progress	Medium
09	Record customer inquiries	Medium
10	Assign and manage sales leads	Medium
11	Approve, reject, or reschedule viewing appointments	High
12	Convert an approved reservation into a sale	High
13	Add and update apartment projects	High
14	Add apartment units with price, size, floor, rooms, and facilities	High
15	Update apartment status as available, reserved, sold, or temporarily blocked	High
16	Upload images, floor plans, brochures, and virtual-tour links	Medium
17	View customer payment submissions	High
18	Approve or reject payment evidence	High
19	Create and manage installment plans	Medium
20	Generate receipts and financial reports	Medium
21	Manage customer and staff accounts	High
22	Assign roles and permissions	High
23	Monitor apartments, reservations, appointments, payments, and sales	Medium
24	Generate system reports	Low

Sprint Planning

- ## Sprint 1 - Foundation and Apartment Listings (Weeks 1-3)

Sprint Goal: Establish secure user access and deliver the core apartment listing experience.

Sprint ID	User Story ID	Selected User Story	Priority
SP1-01	04	Customer registration	High
SP1-02	21	User account management	High
SP1-03	22	Roles and permissions	High
SP1-04	13	Apartment project management	High
SP1-05	14	Apartment unit management	High
SP1-06	01	Browse apartment listings	High
SP1-07	02	Search and filter apartments	High
SP1-08	03	View apartment details	High

- ## Sprint 2 - Appointments and Reservations (Weeks 4-6)

Sprint Goal: Enable customers and staff to manage inquiries, viewing appointments, and apartment reservations.

Sprint ID	User Story ID	Selected User Story	Priority
SP2-01	05	Book viewing appointment	High
SP2-02	11	Manage viewing appointments	High
SP2-03	06	Submit apartment reservation	High
SP2-04	15	Update apartment availability status	High
SP2-05	09	Record customer inquiries	Medium
SP2-06	10	Manage sales leads	Medium

- ## Sprint 3 - Payments and Sales (Weeks 7-9)

Sprint Goal: Complete payment verification, installment tracking, receipt generation, and conversion of reservations into sales.

Sprint ID	User Story ID	Selected User Story	Priority
SP3-01	07	Upload payment evidence	High
SP3-02	17	View payment submissions	High
SP3-03	18	Approve or reject payment evidence	High
SP3-04	19	Manage installment plans	Medium
SP3-05	12	Convert reservation into sale	High
SP3-06	20	Generate receipts and financial reports	Medium

- ## Sprint 4 - Customer Support, Reports and Completion (Weeks 10-12)

Sprint Goal: Complete dashboards, media management, reporting, integration, testing, and final improvements.

Sprint ID	User Story ID	Selected User Story	Priority
SP4-01	08	Customer activity dashboard	Medium
SP4-02	16	Upload apartment media and documents	Medium
SP4-03	23	Administrator system monitoring	Medium
SP4-04	24	Generate system reports	Low
SP4-05	20	Finalize financial reports	Medium
SP4-06	03	System-wide UI improvements and testing	High

Sprint Backlog and Task Assignments

The following sprint backlogs break selected user stories into implementable tasks with estimated hours and clear ownership. The estimates are planning values and may be adjusted during daily Scrum meetings.

7.1 Sprint 1 Backlog

Task ID	Task	Est. Hours	Owner	Related Story
T-01.1	Design customer registration and login interfaces	5	IT25100393	US-04
T-01.2	Implement registration and secure authentication APIs	8	IT25100393	US-04
T-01.3	Validate registration inputs and display messages	3	IT25103871	US-04
T-01.4	Test customer registration and login flow	3	IT25102539	US-04
T-02.1	Design user and staff management interface	5	IT25103871	US-21
T-02.2	Implement user search, status update, and account control APIs	8	IT25103131	US-21
T-02.3	Implement role assignment and permission rules	6	IT25100393	US-22
T-02.4	Test role-based access for all user types	4	IT25102539	US-22
T-03.1	Design project and apartment unit management forms	6	IT25103860	US-13/US-14
T-03.2	Create backend APIs for projects and apartment units	10	IT25103860	US-13/US-14
T-03.3	Create database entities and relationships for listings	6	IT25103860	US-13/US-14
T-03.4	Implement apartment status and validation rules	4	IT25103887	US-14
T-04.1	Design apartment browsing and details pages	6	IT25103860	US-01/US-03
T-04.2	Implement apartment search and filter API	8	IT25103860	US-02
T-04.3	Add filters for price, location, bedrooms, and availability	5	IT25103871	US-02
T-04.4	Test browsing, details, search, and filters	4	IT25102539	US-01/US-02/US-03

7.2 Sprint 2 Backlog

Task ID	Task	Est. Hours	Owner	Related Story
T-05.1	Design viewing appointment request form and schedule view	5	IT25103887	US-05
T-05.2	Implement appointment creation and available time-slot API	8	IT25103887	US-05
T-05.3	Implement appointment approval, rejection, and rescheduling	7	IT25103887	US-11
T-05.4	Test customer and staff appointment workflows	4	IT25102539	US-05/US-11
T-06.1	Design apartment reservation request interface	5	IT25103887	US-06
T-06.2	Implement reservation creation and availability verification	9	IT25103887	US-06
T-06.3	Set reservation expiry date and reservation status	4	IT25103887	US-06
T-06.4	Automatically update apartment status after reservation	5	IT25103860	US-15
T-06.5	Test duplicate reservation prevention	4	IT25102539	US-06/US-15
T-07.1	Design inquiry and lead management interface	5	IT25103131	US-09/US-10
T-07.2	Implement inquiry recording and lead assignment APIs	8	IT25103131	US-09/US-10
T-07.3	Implement follow-up notes and lead status updates	5	IT25103131	US-10
T-07.4	Test inquiry and lead management flow	3	IT25103871	US-09/US-10

7.3 Sprint 3 Backlog

Task ID	Task	Est. Hours	Owner	Related Story
T-08.1	Design customer payment submission interface	5	IT25100393	US-07
T-08.2	Implement payment evidence upload and secure file storage	8	IT25100393	US-07
T-08.3	Validate payment amount, method, and reservation reference	4	IT25100393	US-07

T-08.4	Test payment submission flow	3	IT25102539	US-07
T-09.1	Design finance payment review dashboard	5	IT25100393	US-17
T-09.2	Implement payment approval and rejection APIs	8	IT25100393	US-18
T-09.3	Create transaction history and audit record	5	IT25103131	US-18
T-09.4	Test valid, invalid, and duplicate payment cases	4	IT25102539	US-17/US-18
T-10.1	Design installment plan creation form	5	IT25100393	US-19
T-10.2	Implement installment schedule, due dates, and balances	9	IT25100393	US-19
T-10.3	Calculate paid, pending, partially paid, and overdue statuses	5	IT25100393	US-19
T-10.4	Test installment calculations	4	IT25103871	US-19
T-11.1	Implement reservation-to-sale conversion workflow	8	IT25103131	US-12
T-11.2	Create sale record and link customer, apartment, and payments	7	IT25103131	US-12
T-11.3	Update apartment status from Reserved to Sold	4	IT25103860	US-12
T-11.4	Generate downloadable payment receipt	5	IT25100393	US-20
T-11.5	Test complete payment-to-sale workflow	5	All Members	US-12/US-20

7.4 Sprint 4 Backlog

Task ID	Task	Est. Hours	Owner	Related Story
T-12.1	Design customer dashboard for appointments, reservations, and payments	6	IT25102539	US-08
T-12.2	Implement customer activity summary APIs	7	IT25103131	US-08
T-12.3	Test customer dashboard data and access control	4	IT25102539	US-08
T-13.1	Design apartment media upload and gallery interfaces	5	IT25103860	US-16
T-13.2	Implement image, floor plan, brochure, and virtual-tour management	8	IT25103860	US-16
T-13.3	Test file type, size, replacement, and deletion functions	4	IT25102539	US-16
T-14.1	Design administrator monitoring dashboard	6	IT25103871	US-23
T-14.2	Implement summary counts for users, apartments, reservations, payments, and sales	8	IT25103131	US-23
T-14.3	Implement system report filters and export-ready data	8	IT25103131	US-24
T-14.4	Create financial and sales report queries	7	IT25100393	US-20/US-24
T-15.1	Integrate all frontend and backend modules	8	All Members	All
T-15.2	Perform functional and role-based access testing	7	All Members	All
T-15.3	Fix defects and improve usability	8	All Members	All
T-15.4	Prepare final demonstration and documentation	6	IT25100393 / IT25103871	All